From Panic to Protocol: A Comparative Analysis of the 2008 MMF Run and the 2022 Terra-Luna Collapse

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Executive Summary

Vision

Future-proofing digital markets: Leveraging data-driven insights to reinforce systemic resilience, prevent algorithmic contagion, and ensure financial continuity

Question

Compare the onset and propagation of panic in the 2008 Reserve Primary Fund and 2022 Terra-Luna runs.

Track capital flows in 2008 and 2022 to identify safe-haven assets during crises.

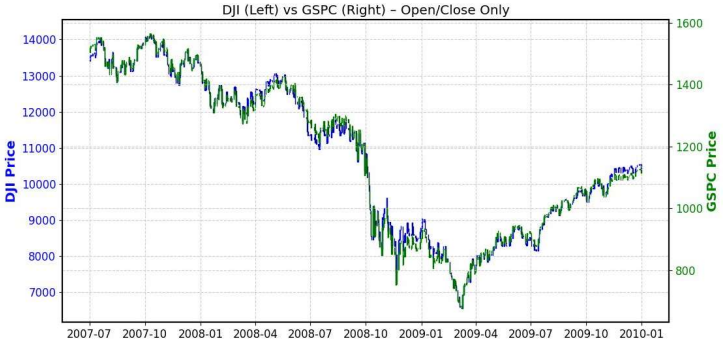
Policy

Stablecoin Reserve Transparency & Audit Requirements

Impact

Trust & Reduced Bank-Run Risk, Market Discipline & Financial Stability

Reserve Primary Fund Run



S&P500 (GSPC) & Dow Jones Index (DJI)
Measures of stock performance for the top 500 and 30 US firms respectively in terms of size

Observation:

Market peaked in October 2007, followed by a gradual decline till Sep 2008 accompanied by a sharp drop to an all-time low on March 2009, before beginning a slow recovery.

Explanation:

GSPC/DJI declines reflected a systemic repricing of financial solvency. Heavily weighted toward banks, these indices signaled a profound collapse in trust regarding counterparty risk and balance sheet viability.

Key Takeaways:

Institutional Trust Index: GSPC/DJI acted as a barometer for systemic confidence.
Pricing Shift: Markets shifted from valuing future cash flows to pricing in default and intervention.
Systemic Signal: High volatility signaled the transition from liquidity shock to full systemic crisis.

Early crisis
Start of a prolonged downturn as credit loss and housing stress mounted. Modest sell-offs reflected rising fears about banking exposures and tightening credit conditions. Indices declined by nearly 20% as confidence dipped.

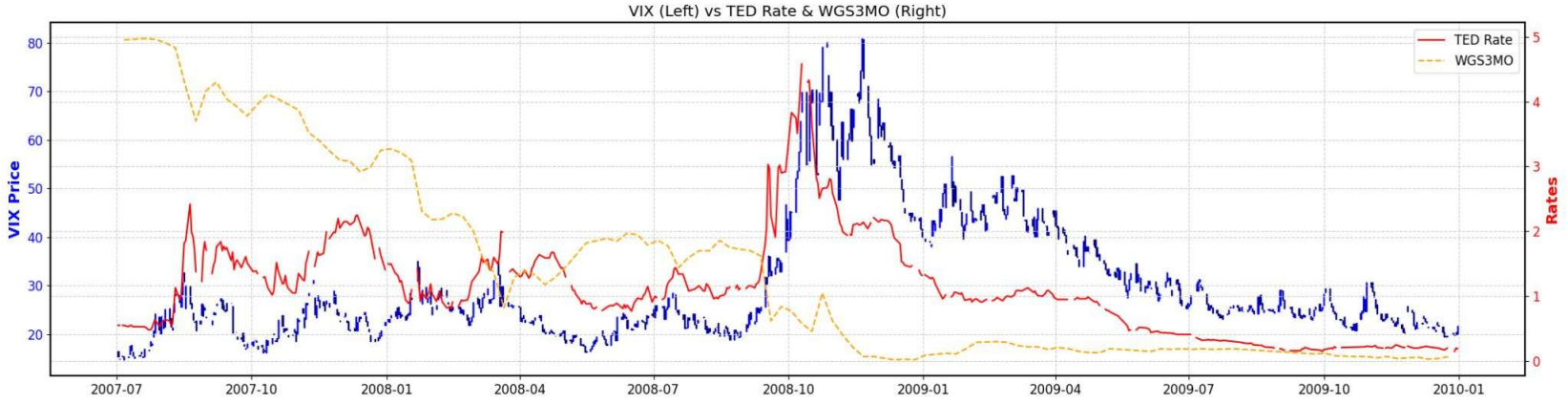
Oct 2007 — Sep 2008 — Dec 2008 — Mar 2009

Lehman shock phase
Markets plunged sharply amid Lehman's collapse and systemic credit freezes. Bailout uncertainty spiked and liquidity dried up as indices fell by more than 35%.

Government intervention
Fiscal and monetary interventions allowed for a partial plateau but economic downturn fears intensified and the decline resumed, with indices dropping to about half its peaked values from 2007.

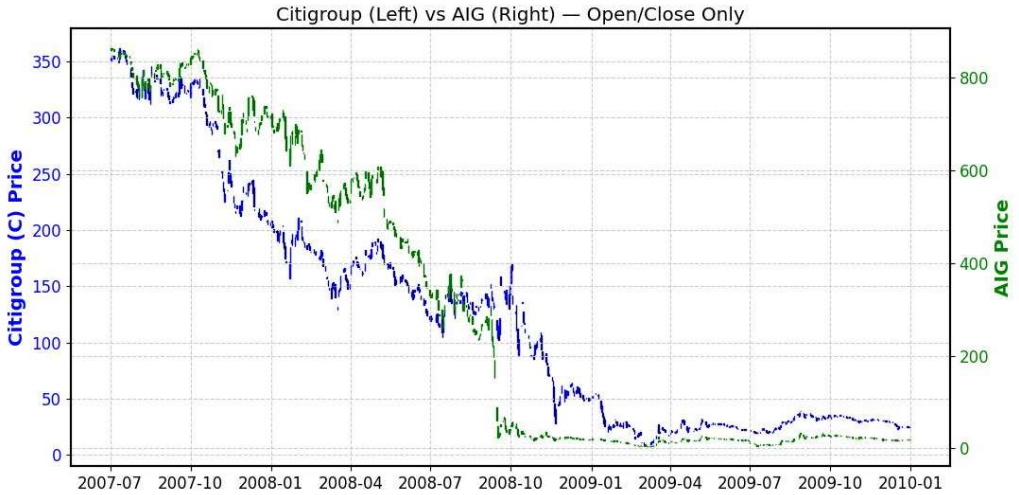
Stabilization phase
Both indices began a sustained recovery as quantitative easing, fiscal stimulus, and improved macro data restored market confidence.

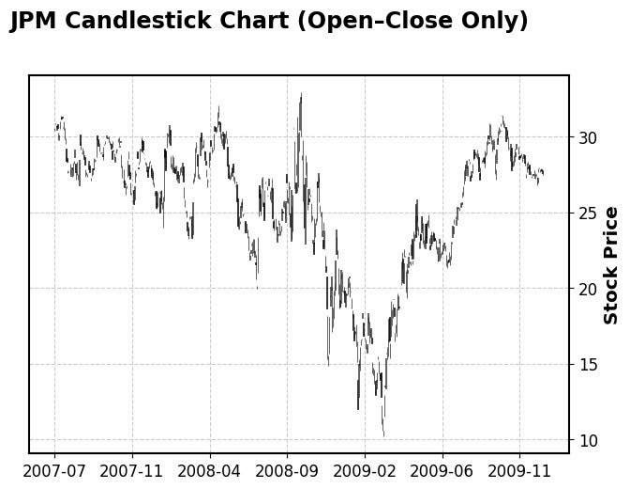
Reserve Primary Fund Run



	VIX **Measures implied volatility in equity options**	**TED Rate** **Measures risk levels perceived between banks**	**WGS3MO** **Captures volatility in 3-month Treasury yields**
Observation:	VIX rose sharply on Aug 2008, shooting past the 20% stability mark and peaking at almost 4x that on Sep 2008.	TED Rate spiked to around 4-5% on Sep 2008, a tenfold rise to pre-crisis levels.	WGS3MO rate collapsed below 1% after Sep 2008.
Explanation:	Signalled extreme volatility. Market participants were pricing extreme uncertainty about future losses. Surging VIX made it more expensive to hold stocks, reinforcing selling pressure and deepening panic.	As risks and returns became uncertain, banks freezed interbank lending. Distrust between financial institutions and counterparties. Liquidity hoarding.	Capital flowed into T-bills as investors treated to **Treasury bills as a safe haven**, driving up demand and lowering yields amid peak uncertainty.

Reserve Primary Fund Run

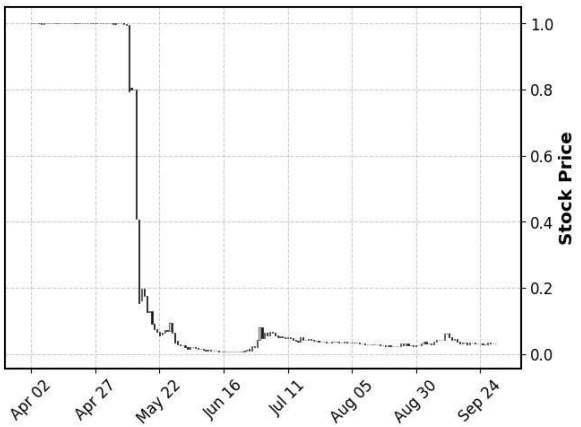




	AIG (American International Group Inc)	C (Citigroup Inc)	JPM (JPMorgan Chase & Co)
Observation:	Steady fall in stock prices since Jul 2007 Steep drop on Sep 2008, coinciding with the general drop in VIX and TED rate Both nearly bottomed out		Price increased in volatility since Jul 2007 Dropped threefold from Sep 2008 to Feb 2009 Quick recovery after Feb 2009
Explanation:	Insurer of global financial system AIG's contracts turned to AIG for collateral Insufficient liquid cash for payouts Confidence in AIG crashed	Large withdrawals of deposits **(VIX)** Limited interbank lending **(TedRate)** **AIG** unable to insure Insufficient liquid funds Confidence in CITI crashed	Operating system of the financial system State chosen stabiliser 1) Received implicit guarantees 2) Emergency liquidity 3) Privileged access to failing institutions Confidence in JPM recovered quickly

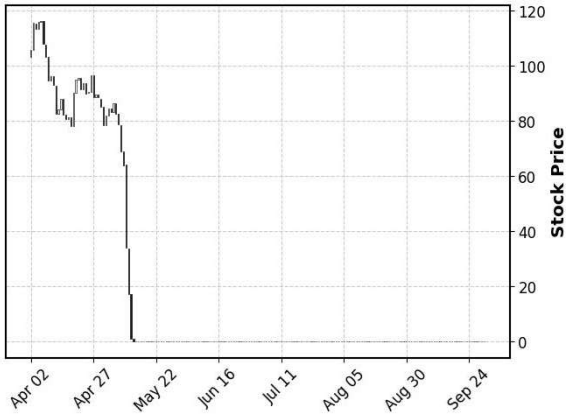
Terra-Luna Stablecoin Run

USTC Candlestick Chart (Apr–Sep 2022)



USTC: Algorithmic stablecoin designed to maintain a $1 peg

WLUNA Candlestick Chart (Apr–Sep 2022)



WLUNA: Volatile cryptocurrency used to back that peg through a mint-and-burn mechanism.

Observation:

De-pegging: Beginning around May 7, 2022, USTC began deviating significantly from its $1 peg, falling to $0.80 before collapsing near $0.
Collateral Crash: Simultaneously, LUNA (or wrapped LUNA) prices crashed from approximately $80 to near zero.
Surge in Volume: Trading volume spiked sharply as investors rushed to redeem USTC or swap for other assets.

Explanation:

Panic-driven redemptions triggered massive LUNA minting via the USTC burn mechanism, causing an oversupply that crashed LUNA's price and created a fatal feedback loop known as a "death spiral".

Pre-stress accumulation
USTC adoption expands rapidly, driven by the Anchor Protocol's ~20% yield and extensive use of USTC as collateral across DeFi. Stability is maintained by confidence in the algorithmic mint–burn mechanism rather than external reserves.

Late 2021 – March 2022

Early–Mid April 2022

Early Warning Signals
WLUNA exhibits rising price volatility, indicating weakening confidence in the backing asset. At the same time, transaction volumes in alternative stablecoins—especially USDC—begin to increase, suggesting anticipatory capital reallocation before any visible peg failure.

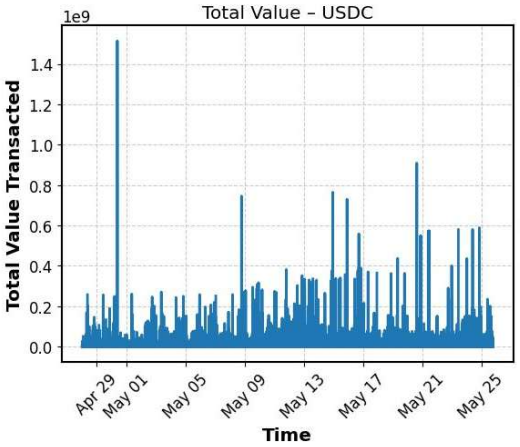
May 9–13, 2022

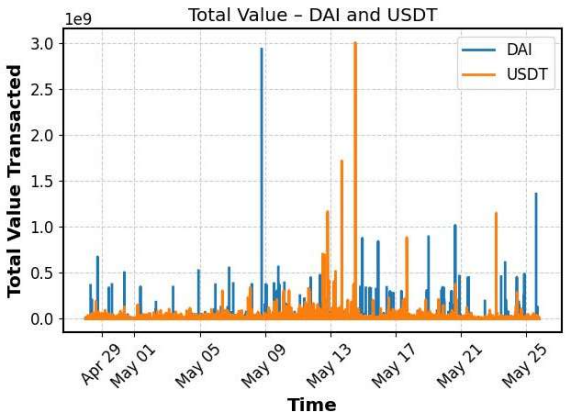
Run and Collapse
Large-scale USTC redemptions trigger the minting of massive quantities of WLUNA. WLUNA's price collapses, undermining USTC's backing mechanism. USTC loses its peg, and transaction volumes in USDC, DAI, PAX, and USDT spike sharply, reflecting panic-driven flight to perceived safer stablecoins.

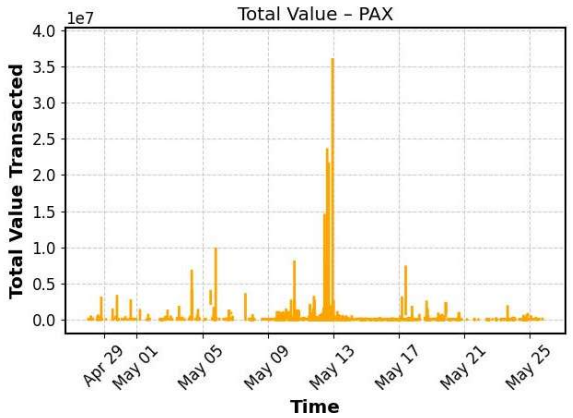
Mid–Late May 2022

Aftermath and Contagion
USTC and WLUNA prices approach zero, erasing tens of billions in market value. Losses propagate to leveraged crypto firms and lending platforms, leading to insolvencies and triggering intensified regulatory scrutiny of stablecoin design and redemption mechanisms.

Terra-Luna Stablecoin Run







Observation:

Spiking volume and price volatility in WLUNA in mid-April indicate that investors lost confidence in USTC's backing mechanism, and began reallocating early into USDC well before the peg failure in May.

Volume spikes in USDT, DAI, and PAX from May 9–13 correlate directly with the USTC peg failure, signaling a panic-driven rush by investors to reallocate funds into recognized stablecoins **not** backed by algorithms but instead backed by fiat reserves or crypto assets.

Explanation:

Investors rushed into **credible stablecoins backed by tangible, liquid reserves rather than algorithms**, causing extreme volume spikes that verified a classic flight-to-safety scenario within the crypto ecosystem. Rather than exiting crypto entirely, investors revealed a **hierarchy of trust**, favoring assets with clearer backing and redemption credibility over those backed by algorithms and confidence when financial stress intensified.

Stablecoin Reserve Transparency & Audit Requirements

Require all stablecoins to publish transparent, independent audits showing actual reserve holdings in real-time or on a reliable schedule.

This can be achieved by:

- **Regular independent third-party audits** by licensed accounting firms
- **Clear disclosure of reserve composition** (e.g., cash, Treasury bills, commercial paper, corporate bonds, crypto assets)
- **Frequent reporting** – ideally real-time dashboards, or at minimum daily/weekly updates
- **Segregation of client assets** so reserves cannot be mixed with the issuer's own funds
- **Legal rights of redemption** guaranteeing that each token can be exchanged 1-for-1 for fiat

Impact

1. Trust & Reduced Bank-Run Risk
    a. Greater confidence in backing
    b. Less panic-driven withdrawals
2. Market Discipline
    a. Risky reserves exposed
    b. Pressure to hold safe, liquid assets (cash, T-bills)
3. Financial Stability
    a. Lower contagion risk
    b. Clear legal protection for holders
    c. More institutional adoption

Conclusion